

Molecular Dissection of Hairpin Ribozyme Catalyzed Self-Cleavage

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Abstract

[...]

Introduction

The discovery of catalytic RNAs, or ribozymes, drastically changed our understanding of biological catalysis by demonstrating that cellular activities are not exclusively performed by proteins. However, the detailed chemical mechanisms underlying ribozyme catalysis remain uncertain, with ongoing debates regarding how these catalysts facilitate their reactions.

Ettoffer un peu ici. Peut être faire un petit inventaire de ce que l'on connait des autres ryb ?

Among the most studied ribozymes, the hairpin ribozyme (HpR) stands out for its ability to cleave a phosphodiester bond within its own backbone, replacing a 3'-5' polymeric linkage with a 2'-3' cyclic terminating linkage.^{1,2} Although HpR is not the only self-reactive ribozyme, it is one of the few capable of catalyzing this reaction without the requirement for divalent metal ions.³⁻⁷ Although it is established that all chemical components required

for the self-splicing reaction are part of the RNA itself, we do not know exactly which RNA atoms take part, nor do we know exactly how they proceed with the self-cleavage. The central step of this transphosphoesterification process is an addition–elimination reaction that involves formation of a pentavalent phosphorus transition state and requires precise alignment of the nucleophilic group (O2'), the central phosphorus atom, and the leaving group (O5'). Building from there, two key questions arise: (i) how is the nucleophilic group activated prior to the addition step? and (ii) how is the departing group activated prior to the elimination step?

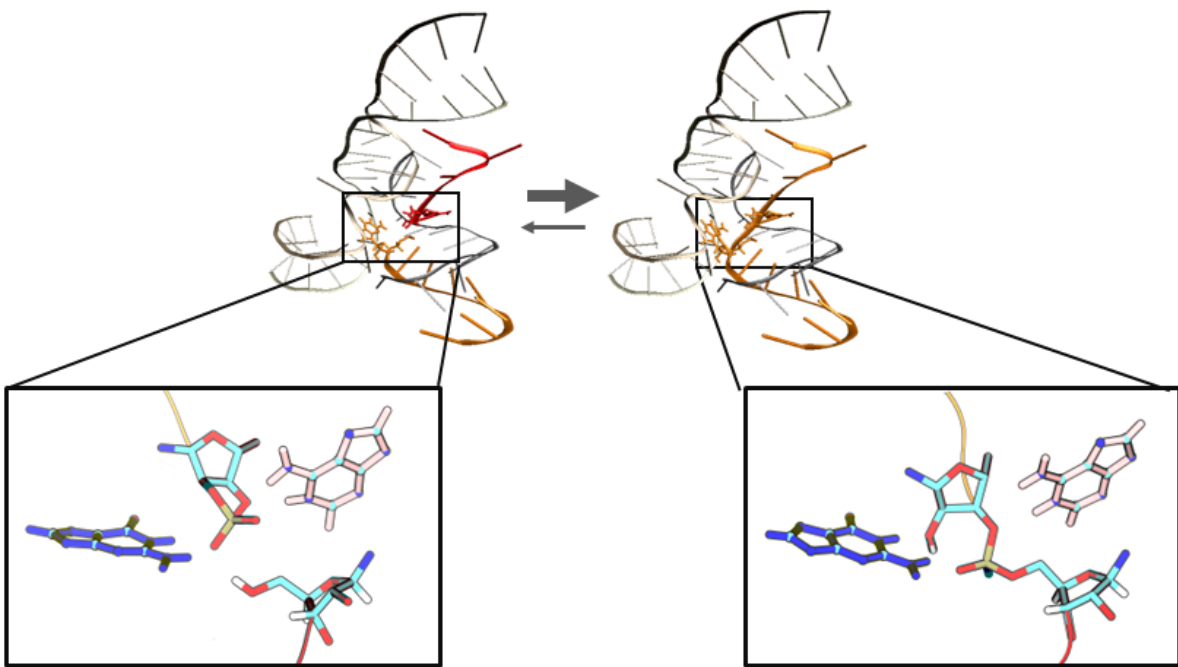
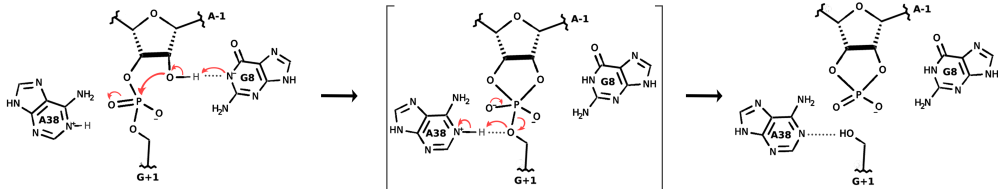


Figure 1: **TEMPORARY FIGURE AND DESCRIPTION** Schematic representation of the hairpin ribozyme self-cleavage reaction.

Mutagenesis experiments,⁸ reinforced by structural resolution studies,^{9–12} have demonstrated that two surrounding nucleobases, G8 and A38, are mandatory for the chemical activity and are in the direct neighbourhood of the A-1 — G+1 junction. From there, a first mechanistic scenario was proposed:¹³ the dianionic scenario proposes that both G8 and A38 are directly involved, acting respectively as the general base that activates the nucle-

ophile O2' by capturing its proton, and as the general acid that reprotonates the leaving group O5' after the elimination step (see Figure 2 A).¹³ Despite supporting pH-dependency experiments^{14,15} and matching transition state arrangements,¹⁶ this cleavage scenario suffers from one major inconsistency, which is the necessary prior deprotonation but *in vivo* impracticality of the N1 atom of the G8 nucleobase.¹⁷

General Base/ General Acid Dianionic Scenario



Inline Monoanionic Scenario

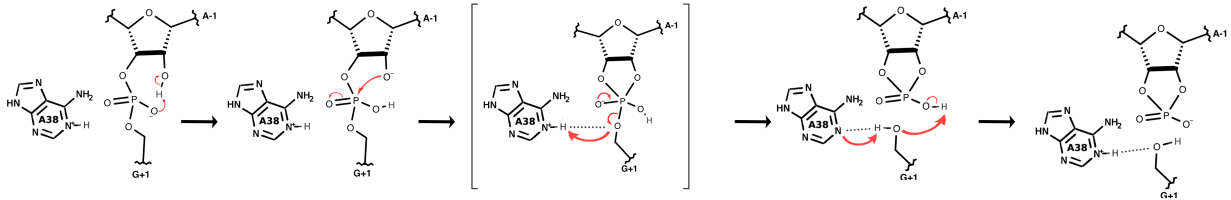


Figure 2: Detailed schemes of the three proposed mechanisms for the Hairpin Ribozyme self-cleavage reaction: the general acid-base mechanism, the inline mono-anionic mechanism, and the A38 mediated proton shuttle scenario. We chose here to represent the inline mono-anionic mechanism with the A38 mediated proton transfer in the elimination step.

To address this insufficiency, a second scenario has been considered:¹⁸ the in-line monoanionic scenario (see Figure 2 B). Here, A38 and G8 nucleobases only contribute to the electrostatic stabilization of the transition state, being involved in several H-bonds with atoms of the scissible junction. These non-covalent interactions force the catalytic site to arrange favorably for the initiation of the reaction. While being maintained in position, the deprotonation of the O2' atom of A-1 is performed by one of the non-bridging oxygens the phosphate group itself, leading to a monoprotic phosphate intermediate. However, this scenario also suffers from its apparent incompatibility with the experimental pH-dependency of the reaction, as argued by Kath-Schorr et al.¹⁵

All this debate remains unsettled so far because of the inherent self-reactiveness of the

HpR. Indeed, because it is self-reactive, the hairpin ribozyme can not be structurally resolved in directly as such. Instead, mimics of the HpR in which the chemical activity is inhibited, either by removing the O2' oxygen nucleophile, by methylating it or by replacing other chemical groups with non-reactive ones, are used to study the catalytic site.^{9,10} Thus, no one can directly observe the catalytic site in its reactive form. With arguments based on crystal structures of *mimics* of the catalytic states, and non-discriminative pH profiles, it has been hard to draw definitive conclusions from experiments only.

In parallel to the experimental efforts, the HpR has been studied using *in silico* tools. Complementary to X-ray crystal studies, molecular dynamics (MD) expands the knowledge gained through structural resolution by providing the dynamic behavior of the resolved object. Here, the challenge of studying self-reactive objects is no longer a problem, as one can simulate any reactive species without worrying about self-degradation. Computational studies of the HpR are of two sorts. First, extended simulation of the full HpR with classical force fields,^{19–21} which allow for the exploration of the conformational landscape of the HpR.^{19–21} Second, combined quantum and classic mechanics (QM/MM) simulations,^{22–24,24,25} which investigate the chemical activity of the catalytic site by simulating the chemical steps of the reaction. These two investigations means are usually conducted sequentially, as QM/MM simulation require relevant pre-catalytic initial structures which can only be obtained from prior robust classical MD simulations.

The contributions of the classical MD studies can be classified into two opposing pitches. The first base itself on study of the catalytic site which has been explicitly constrained the catalytic site elements to ensure the formation of a specific H-bond network which are coherent with the general dianionic scenario: G8:N1...A-1:O2', A38:N1...G+1:O5', G8:N2...G+1:Opro-Rp, and A38:N6...G+1:Opro-Rp.^{20,26} Although a certain stability of this local conformation was evaluated within 500 ns of simulation, we proved in our previous work that the stability of this local arrangement, which we labeled L₁, does not hold true over extensive sampling.²⁷ In fact, we rather found that another local arrangement, L₂, is the true stable conformation of

the hairpin ribozyme in its pre-catalytic state. These observations are in line with the second pitch which stipulates that this L₂ arrangement, in which G8 is protonated and distinct from O2' and O2' involved in a H-bond with the non-bridging pro-Rp oxygen, could be the departure point of the in-line monoanionic scenario. Kumar et al. explored the conformational landscape of the catalytic site once the O2' proton has been transferred to Opro-Rp, and showed that the catalytic site followingly arranges itself perfectly for the upcoming addition-elimination steps, with the O2'-P-O5' inline attack angle IAA flatter than 140°. ²¹ [Il manque la référence a d'autres articles ou ces états intermédiaires on été étudiés comme](#)

In parallel to the classical MD simulations, several efforts have been made to investigate the chemical activity of the catalytic site using QM/MM simulations. [\[...\] Allonger sur la QM/MM ?](#)

Far from settling the debate, all these *in silico* studies have rather highlighted the complexity of the catalytic site conformational landscape, and the difficulty to draw conclusions given the inherent limitations of MD, and the limited conformational sampling of QM/MM simulations.

In this study, we propose to complement the arsenal of *in silico* studies which have conducted on the HpR and to explore the conformational landscape of the different intermediates of the two opposed scenarios within a comparative study, and using state of the art extensive sampling techniques, Replica Exchange with Solute Tempering or REST2, which have proven to be essential in the study of systems as flexible as the HpR. ²⁷ This will lead us to yet so far undone simulation of the pro-Sp monoprotic HpR, and several cleaved states of the HpR. With this, we provide valuable insights into the mechanistic debate, by addressing the feasibility of the mechanistic steps at each stage of the reaction. Additionally, these unprecedented simulations provide relevant equilibrated structures along the different mechanistic routes, an sampling effort which will be useful, if not mandatory, for subsequent quantum calculations.

We find that not only does the pre-catalytic catalytic site of the HpR arrange coherently

with a inline-monoanionic scenario, but it also adopt flat IAAs ounce the proton has been transfered on one of the two non-bridging oxygens, favorably with the addition elimination next step. This favorable behavior appears both for the pro-Sp and the pro-Rp pathways, opening the door to two possible inline monoanionic sub-scenarios. On the contrary, our efforts do not support the general acid-base mechanism, as first the deprotonation of G8 as required for this nucleobase to act as general base drives it away from the scissible junction, and secondly, the proton transfer from the O2' to G8 creates a charge on O2' which seem repulsed from the phosphate group, making it impossible for the nucleophilic attack to occur.

Manque une phrase bilan.

Materials and Methods

Simulation parameters and Analysis Softwares

The different simulations of the hairpin ribozyme intermediates were performed using the GROMACS 2022.3²⁸ patched with PLUMED 2.7.^{29,30} The visualization and analysis of the REST2 trajectories were performed using respectively the VMD software, with the latest COLVAR extension, and several home-made scripts (available on the Github repository of the group) which require both the MDAnalysis and the Baranaba Python libraries.

Forcefields. Coherently to the conclusions drawn in our previous work,²⁷ we parametrize all our systems with the Amber14_ff99bsc0χ_{OL3}ϵζ_{OL1} force field.³¹ This force fields does not include the parameters for certain nucleobases present in the HpR intermediates studied in this study and had to be specifically added. The parameters of A38 N1 protonated adenine, the joined adenine and guanine forming the monoprotic phosphate protonated on its Opro-Rp (GAA) and deprotonated on the adenine O2'(RAA) were extracted from the literature²⁷ and adapted to the forcefield manually. The parameters for the monoprotic phosphate group with the proton on the pro-Sp atom (labeled RSA + GSA) were adapted from these very

RAA and GAA parameters by simply inverting the charge of pro-Rp and pro-Sp oxygens, without modifying the rest of the redistribution. Both the O2' deprotonated A-1 used in the dianionic scenario (AD) and the 2',3'-cyclic phosphate adenine used in the two cleaved states (RAC) were generated using Gaussian³² and Ambertools³³ softwares to obtain the charge redistribution and generate appropriate new atom types. All parameters can be found in the supporting information.

Initial configurations. We chose to initiate the simulations of the ligated dianionic scenario intermediates from the crystallographic structure of the pre-catalytic state of the minimal HpR (PDB ID: 2OUE),⁹ with methyl group present on the O2' atom of A-1 was manually removed, coherently with what we for the simulation of the inline monoaionic pre-catalytic state previous work.²⁷ The pro-Rp monoprotic intermediate HpR simulation was initiated from a L2 conformation (see the descriptors section for details about the conformational labelling) extracted from this very pre-catalytic state simulation. As initial structure for the pro-Sp intermediates, we used a L_1^{pucker} frame from the trajectory provided by Mlynsky et al. (see our previous work for conformational labelling²⁷ for more details on this simulation).

The initial structures of all the cleaved states investigated in this study were constructed from the crystallographic structure of the 61-mer minimal hairpin ribozyme resolved at 2.2 Å, in which the G+1 phosphate group was removed (PDB ID: 2P7D).³⁴ The missing phosphate (P, Opro-Rp, Opro-Sp atoms) was positioned according to the Vanadate Structure (PDB ID: 2P7E). After superposing the respective A-1 nucleotides, the Cartesian coordinates of the vanadate were extracted to be used for the phosphorus atom, and the same goes for the oxygen atoms.

Additionally, a conformational heterogeneity at nucleobase G8 in the cleaved product mimic crystal structure was observed, with 50% of the population with G8 in the syn conformation, 50% in the anti conformation.³⁴ Both conformations were reported in the crystal structure. We chose to keep the anti conformer, as this is the conformation of the ligated structure, the

most populated conformer of the vanadate transition state mimic (PDB ID: 2P7E), and the one that was resolved in the pre-catalytic state (PDB ID: 2OUE) and in the product state of the 4WJ ribozyme (PDB ID: 1M5K). The positions of the final catalytic site atoms (in the crystal structure) were compared to the only other cleaved HpR which has been resolved, which differs in sequence and size while conserving the same catalytic core (PDB ID: 1M5V). We demonstrate excellent consistency, with an RMSD of 0.93 Å calculated over all heavy atoms of residues A-1, G+1, G8 and A38 (see Figure 5).

System set up, equilibration and production. All HpR were solvated in TIP3P water using Gromacs `solvate` command. Crystallographic waters were kept, while crystallographic ions were replaced by a 0.2M solution of KCl, and all solvated ribozymes underwent minimization, equilibration, and production runs within the same process as described in our previous work.²⁷ Shortly, the solvent was progressively equilibrated with a 5000 step energy minimization, a 500 ps NVT and a NPT equilibration at $T = 298.15$ K and $P = 1$ bar, all while maintaining the solute with 1000 kJ/mol/nm² positional restraints on the nucleic backbone. The backbone is gradually loosened during three successive relaxations with decreasing constraint forces, from 1000 to 100 to 10 kJ/mol/nm². Finally, the complete system is freely simulated for 100 ps in NVT and 100 ps in NPT terms. The particle-mesh Ewald method³⁵ is used to calculate the electrostatic interactions, and a Verlet cutoff scheme of 10.0 Å is used for Lennard-Jones interactions. Cubic interpolation is used to set the grid size at 0.12 nm. All bonds with hydrogen atoms are subjected to the LINCS algorithm,³⁶ and the simulation timestep was always set at 2 ps. The production might then start from there. Gromacs 2022.3²⁸ patched with PLUMED 2.7^{29,30} was used for all simulations and some system preparation.

Replica Exchange with Solute Tempering (REST2) setup. All HpR simulations were performed using replica exchange with solute tempering (REST2),³⁷ with the same parameters as the ones used in our previous work.²⁷ We employed 24 replicas with rescaling

factors ranging from 1 to 0.667, trying exchanges every 1 ps, at 300 K. The simulations typically lasted for 500 ns, and displayed a average exchange rate of 15—20%.

Descriptors.

We adopted the same conformational labelling as in our previous work²⁷ in order to characterize the arrangements of the catalytic site in its ligated state. This labelling system is entirely based on the combination of observables implying the catalytic site atoms, and distinguish three main conformations. In L_1 , the A-1 residue is in the C2'-endo puckering pseudo-rotation angle, the O2' atom is involved in a H-bond with G8:N1, and the phosphate is orientated such as the O2' is closer to the pro-Sp than the pro-Rp oxygen. In the L_2 conformation, the A-1 residue is in the C3'-endo puckering pseudo-rotation angle, the O2' atom is involved in a H-bond with A38:N6 or with A10:N6, and O2' is closer to the pro-Sp than it is to the pro-Rp oxygen. In the L_C conformation, the A-1 residue is in the C2'-endo puckering pseudo-rotation angle, the O2' atom is involved in a H-bond with G8:N1, and O2' is closer to the pro-Rp than it is to the pro-Sp oxygen. Finally, all frames which do not meet these different conditions are classified as "other" conformations.

Convergence.

The convergence of the simulations of all ligated intermediates was assessed both at the global scale (using the eRMSD quantity as defined in the literature) and at the local scale, using the ratio of frames that fall under the different conformational categories defined above in each time block. Figure ?? A-D details the time evolution of these metrics for the ligated states. The convergence of cleaved states was assessed using only the eRMSD global metric only, as we did not settle any satisfying local collective variable (see Figure ?? E-F).

Results

The notion of "intermediate" commonly used in this study should not be confused with the notion of chemical intermediate state. Here, we only consider the different protonation states of the catalytic site which may appear along the different mechanistic pathways, without any claim on their stability or relevance. Deprived of this knowledge, we choose here to consider accordingly every species that could appear along the different mechanistic pathways, starting with description of certain of the in-line mono-anionic scenario states and ending with those of the general dianionic scenario.

Mono-Anionic Scenario

The mono-anionic mechanism begins with a proton transfer from the O2' to one of the two oxygens of the scissile phosphate, leading to the formation of the "AP" intermediate (see Figure ??). From there, the deprotonated O2' oxygen attacks the phosphorus atom, forming a pentavalent phosphorus transition state. The system subsequently proceeds an elimination step, with the capture of the proton on the phosphorus oxygen and the departure of the substrate group.

The mono-anionic mechanism has two prospects, namely the pro-Sp (O1P) and the pro-Rp (O2P) pathways. Although they are appearing symmetrical on paper, these two pathways are not equivalent in terms of local arrangements and energy. Indeed, they imply different H-bond acceptors for HO2', meaning that they are initiated from conformations of the catalytic site which probably diverge in stability.

In our previous study,²⁷ we have shown how the exploration of the conformational landscape of the pre-catalytic state of the HpR, with A38 protonated on its N1 atom, reveals

we have shown that the pre-catalytic state of the HpR is arranged in a L₁ conformation, We explored accordingly the conformational landscape of both protonated intermediates.

Pre-catalytic state. This matter has already been addressed in our previous study[?] in which we sampled the conformational landscape of the pre-catalytic HpR. We have shown ..., Here again, the question of the relative stability of the L_2 conformation compared to L_1 or to the L_C arrangements proves to be central to this debate. Without any undoubtful arguments, we choose to consider both pro-Sp and pro-Rp accordingly, leaving to the side the remaining question of the instability of the L_1 conformation.

AP intermediates. Starting with the pro-Sp pathway, we sampled the HpR structure in which A-1:O2' is deprotonated, and the pro-Sp oxygen of the scissile phosphate is protonated (see Figure ??).

Once HO2' has been transferred to the pro-Sp oxygen, the nucleophilic O2' is poised to attack through its lone electron pair. The key observables are the inline attack angle (IAA) and $d_{O2'-P}$, which we represented in a 2D distribution plot in Figure 3 A, right. The observation is unequivocal: the catalytic site is arranged perfectly for the nucleophilic attack, with more than 73 % of the frames in the *reactive* region (IAA $\geq 140^\circ$ and $d_{O2'-P} \leq 3.5$ Å). On the other side of the scissile phosphate, A38 is properly positioned and faces the leaving-group O5' oxygen, with 88% of the $d_{O5'-A38:N1} \leq 3.5$ Å.

Additionally, the L_1 is conserved if not reinforced, with 71.0 % of the frames in this conformation during the converged part of the simulation, the rest being unlabeled arrangements. This conclusion is detailed by the catalytic 9-key observables combination analysis which reveals a clear dominance of the L_1 conformation (see 3 A, left, and see Methods of Chapter ?? for more detail regarding this type of analysis).

The combination groups which do not fall into the L_1 criteria are very similar to it, with only the puckering pseudo rotation angle of A-1 shifting away from its C2'-endo range. The 9-key observables combination analysis shows the clear domination of combination groups closely related and highly, with only 9.6% representing rare combinations and more than four combination groups comprising $\geq 5\%$ of the population.

Recent experiments have explored the importance of Watson-Crick G+1—C25 base pair, evaluating the evolution of kinetics and thermodynamics during G+1—C25 breaking.³⁸ A clear correlation emerged between internal equilibrium towards ligation and the increasing number of hydrogen bonds formed between residues +1 and 25. These results suggest a dual role for the G+1-C25 base pair: not only does it stabilize interdomain interactions, but it also constrains the catalytic site to an optimal geometry. This is how the authors interpret the greater sensitivity of the ligation rate to G+1 and C25 mutations, as the precise alignment of the catalytic site should be even more crucial in the direction of ligation, when the 2',3'-cyclic phosphate and 5'-hydroxyl termini are covalently linked.

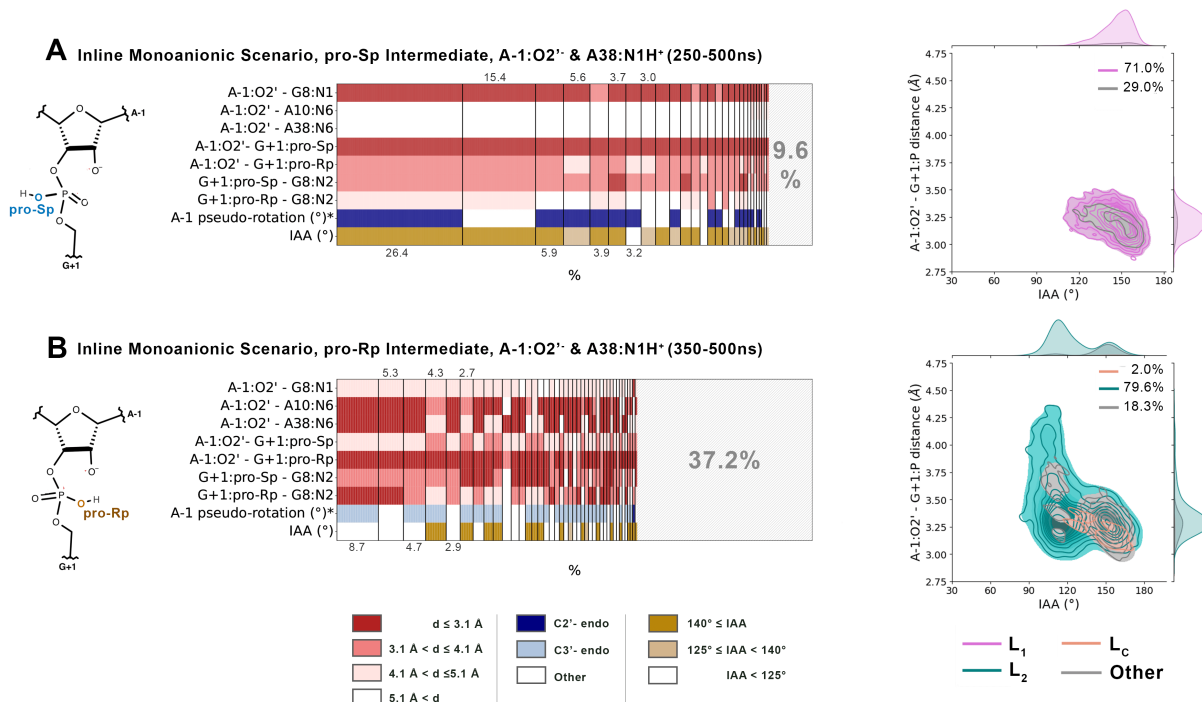


Figure 3: Overview of catalytic site behavior during the converged part of REST2 simulations of the HpR intermediate states in the inline monoanionic scenario: intermediate of the pro-Sp pathway (A), intermediate of the pro-Rp pathway (B). Left: Main combinations of the cleavage pathway key observables (9-key observables combination analysis). Right: 2D distribution of Inline Attack Angle (IAA) and A-1:O2'...G+1:P distance ($d_{\text{O2}'-\text{P}}$). See Methods for more details.

Similarly to the pro-Sp pathway, we have plotted the IAA versus the $d_{O2'-P}$ in Figure 3 B, right. A local minimum, absent in the pre-catalytic state 2D distribution, emerged in the $IAA \simeq 155^\circ$ and $d_{O2'-P} \simeq 3.25 \text{\AA}$ region. These frames are ideally arranged for the nucleophilic attack and represent 35.1 % of the converged REST2 simulation.

Additionally, this region features the rare presence of the L_C conformation, with 3.5% of the $IAA \geq 140^\circ$ frames adopting this arrangement. This caught our attention, as in L_C , the A-1 ribose is in a C2'-endo pucker. This specific puckering angle has been at the center of the discussion of the pre-catalytic state conformational space (Chapter ??). On one hand, most of the crystal structures of the HpR pre-catalytic state display A-1 in C2'-endo. On the other, in all trajectories initiated from the crystal structure, A-1 pucker irrevocably flips from C2'-endo to C3'-endo. Thus, we were intrigued by the presence of the L_C conformation in the pro-Rp intermediate at converged times. When looking closer into the time block analysis of the conformational labeling, we observed a neat diminution of the L_C conformation. Thus, the few frames classified as L_C could be a remnant of the initial conformation, and their eventual disappearance is probable (see Appendices).

Cleaved State

Dianionic Scenario

Discussion

Comparison with Experimental Data

Our results contrast with some of the available experimental data. At the forefront of these experimental elements is the pKa of the phosphate oxygens, which, being around 1, could not easily accept a proton from a neutral A-1:O2'. We note that the pKa of the A-1:O2' atom was re-evaluated to be approximately 18.5,²⁶ making it highly unlikely for either G8 or the phosphate oxygens to capture it. Whether through the pro-Rp or the pro-Sp pathway, the

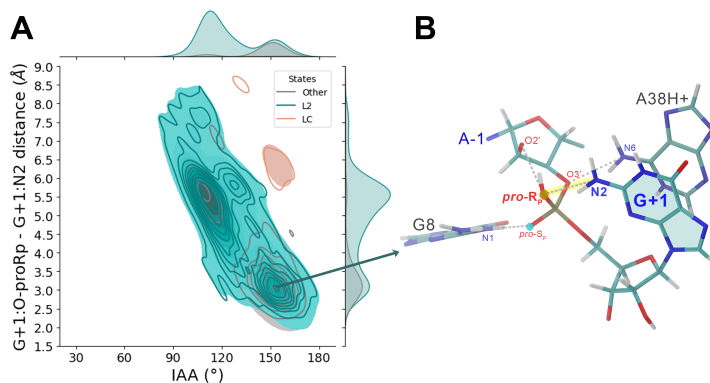


Figure 4: 2D distribution of the distances between protonated Opro-Rp phosphate oxygen and G+1:N2 against the Inline Attack Angle (IAA) in the protonated pro-Rp intermediate of the inline monoanionic scenario (A), together with a 3D view of the catalytic site in the high IAA region (B). The $d_{\text{OproRp-G+1:N2}}$ is highlighted in yellow to emphasize the role of the 2-amino group of G+1 in the stabilization of the IAA flat junction.

monoanionic theoretical pH-profile should exhibit high activity at low pH and a sharp drop around pH 5.5. Experimentally, however, we observe the opposite trend, and no alternative explanation has been proposed to support a monoanionic scenario.

The dianionic scenario on the other hand, has been argued as the compatible with the experimental pH rates.^{15,39} However remains the question of the deprotonation of G8 at *in vivo* pHs. If the G8:H1 proton was to be deprotonated on the fly, then by whom would it be captured? And why would such a chemical specie not directly deprotonate the HO' itself? Our simulations did not show any evidence of unexpected chemical players, and left the question of the deprotonation of G8 unanswered.

Encouragingly, other experimental findings do not directly contradict our monoanionic-favoring perspective. Notably, the crystallographic resolution of the HpR in its transition state analog³⁴ aligns closely with the H-bond network described by the highly stable pro-Sp intermediate, featuring the G8:N2...G+1:Opro-Sp and G8:N1...A-1:O2' H-bonds. Our results are consistent with G8 and A38H⁺ acting as stabilizing agents. Their key roles in terms of electronic stabilization could explain the pH-dependence of the catalytic rate^{13,14,39-42} and the sensitivity of the reaction to the mutations of these residues.^{19,20} They need not be directly involved in catalysis.

It is interesting to note that the monoanionic scenario, with phosphate-assisted proton transfers, is actually reminiscent of the uncatalyzed mechanism in bulk water, that was recently studied by some of our group using deep-neural network inter-atomic potentials in particular,^{43,44} and by others before.^{45,46} The role played by the ribozyme active site residues here would be to provide an electrostatic environment minimizing the free-energy barriers of the reaction,⁴⁷ but this reaction would proceed through a very similar pathway as compared to the bulk.

Our approach has fulfilled its goal, which was to provide a dynamic view at the atomistic level of the ribozyme, when most of the experimental data are rather static and non-discriminative between the structure’s misfolding and chemical inactivity. Nonetheless, we find that the available experimental data remain insufficient to discriminate between the two scenarios. Furthermore, our *in silico* insights have not bridged these gaps; instead, they have highlighted scenarios that seem improbable based on current experimental observations. Importantly, we note that no experiments have been specifically designed to test the monoanionic scenario.

Even if future QM/MM studies validate the enhanced-sampling MD-derived theory of the hairpin ribozyme mechanism, it is essential to acknowledge the experimental data that our results cannot explain, and the settlement of the debate will not be achieved without a solid experimental basis.

Are Equilibrium Molecular Dynamics Simulations Relevant?

The interpretation of our observations should be done with caution.

The approach itself invites scrutiny: intermediate species, by definition, are inherently unstable, and understanding their equilibrium behavior does not necessarily yield meaningful insights into a mechanistic pathway, which is inherently of high energy. This critique is especially pertinent to dianionic intermediates, whose very existence is debated, as the mechanism is often considered concerted.²²⁻²⁴ In the end, distorted and unstable intermediates

are both predictable and expected, offering little insight into the validity of this pathway.

Another critique concerns the force field. Without entering into the debate of the force field accuracy, we will only highlight how force field biases, which may distort the depiction of the conformational landscape, make it impractical to rely solely on conformational sampling to resolve this debate. This sensitivity to parametrization is even more critical when studying intermediates, for which the charge redistribution was determined by association and could easily be questioned.

Building a QM/MM study on the Sampled Intermediates

Despite its limitations, our conformational sampling effort remains indispensable for the setup of a comparative QM/MM study. The mechanistic debate cannot advance without evaluating, within a unified framework, the energy barriers of the different mechanistic pathways to identify the most favorable chemical route.

As noted earlier, this endeavor requires a robust starting point. Yet defining a good starting conformation for ribozyme reactivity studies presents significant challenges, with the critical need for a reliable strategy to characterize the conformational ensemble before undertaking any direct investigation of the chemical steps.

Experimental findings, particularly the pronounced loss of ligation activity resulting from perturbations to the active site architecture, emphasize the importance of precise positioning and orientation in catalytic rate acceleration. Classical MD simulations, despite their inherent limitations, remain central to accessing local arrangements and advancing our understanding of the mechanism. Currently, they provide the only practical means to explore large conformational changes that result in the complete reorientation of key nucleobases. As previously noted, these changes remain beyond the reach of structural resolution techniques, both due to their static nature and the inhibitory strategies employed to observe reactive species like the hairpin ribozyme.

Conclusion

The conformational landscapes of the different mechanistic intermediates examined in this paper have provided a comprehensive understanding of how changes in protonation states lead to complete rearrangements of the catalytic site of the hairpin ribozyme.

Acknowledgement

Supporting Information Available

The overall structure was similar to the equivalent pre-catalytic state used for ligated state simulations (PDB ID: 2OUE), with an RMSD calculated over the nucleic backbone of the 60 common residues evaluated at 0.58 Å. The product crystal structure differed only from the minimal methylated pre-catalytic state by the presence of a 9-triethylene glycol linker (S9L) in place of residue 14, which ensured the junction between H2 and H3. The linker was kept in the simulation.

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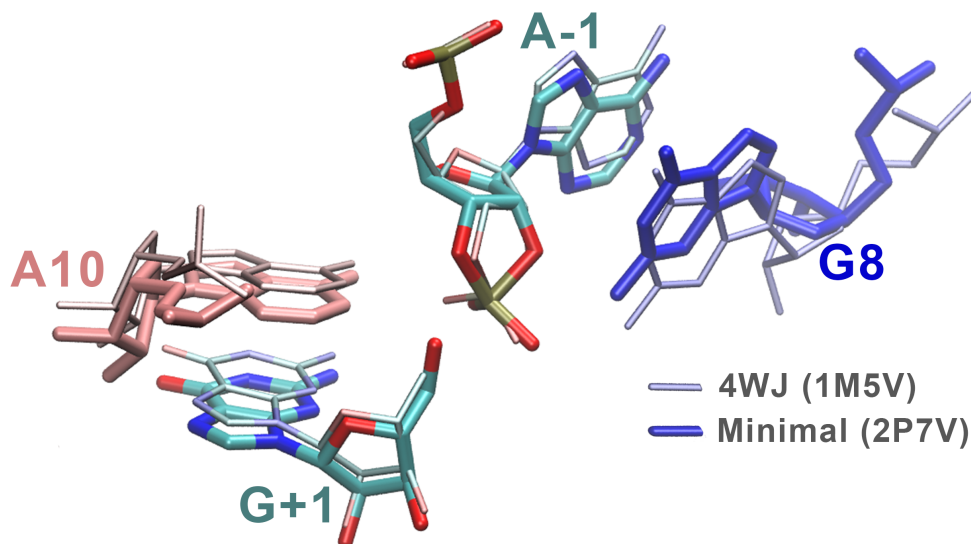


Figure 5: Comparison of the Cleaved States Crystal Structures: the 4WJ hairpin ribozyme (PDB ID: 1M5V) in light colors and thin representation, the minimal hairpin ribozyme (PDB ID: 2P7D) used as initial structure for the cleaved states in this study is strong colors and thick representation. The two catalytic sites differ by only 0.93 Å, calculated over all heavy atoms appearing here, that is to say residues A-1, G+1, G8 and A38.

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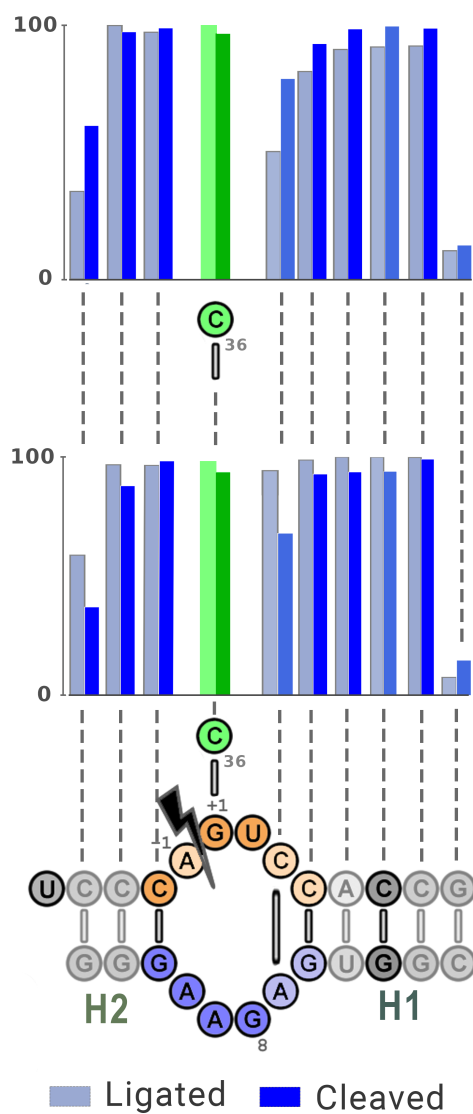
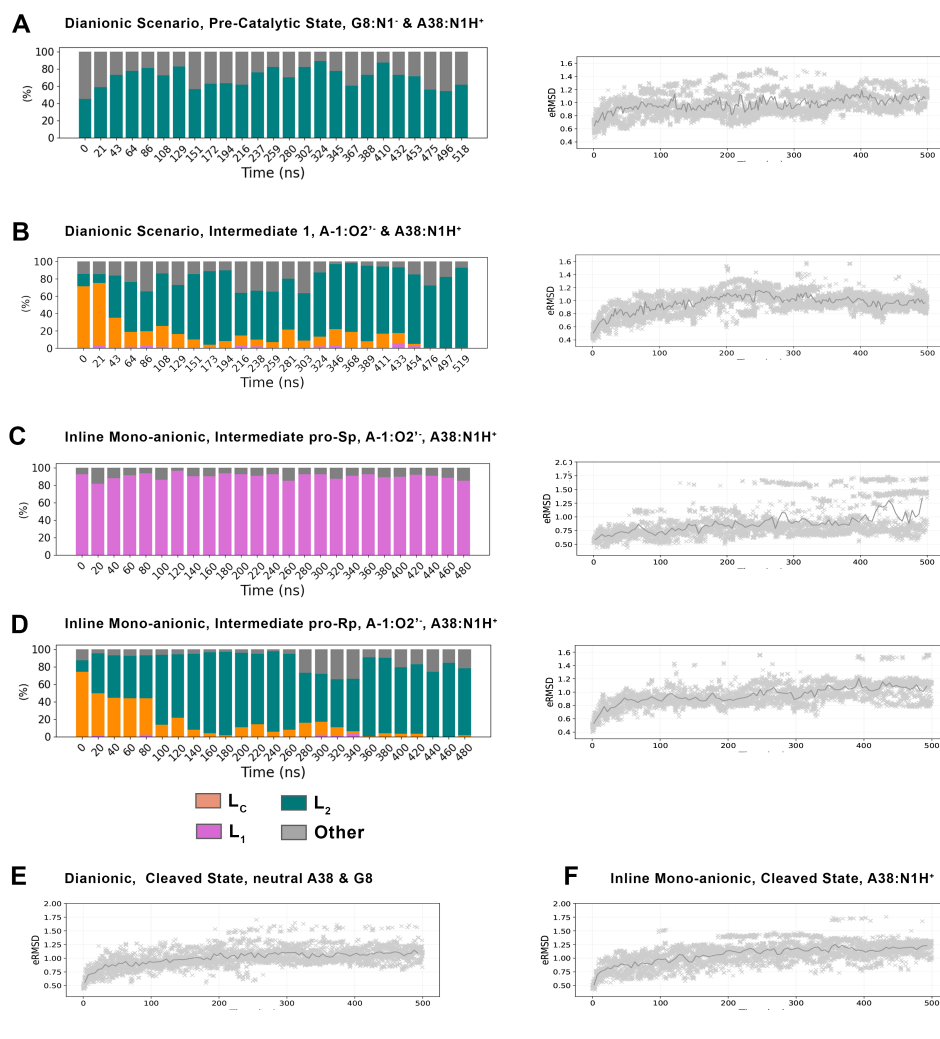


Figure 6: Proportion of the trajectory in which helices H1 and H2 base pairs are formed in the ligated (in pale colors) and cleaved (in strong colors) states of the monoanionic scenario. All Watson-Crick base pairs are evaluated, including the interloop G+1-C25 ribose zipper-member one (in green). The secondary structure is schemed for easing the reading. The scissible phosphate is represented by a black lightning bolt.

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